The logical construction of geometry

Why a geometry course begins with words it refuses to define, and how everything else is built on top of them.

LESSON 1 | 1 × 40 MIN | GEOMETRY 10, §1 | IGCSE E4.1

LESSON CLOCK

8' | 12' | 12' | 8' | 5'

The three undefined terms	8 min
Axiom, definition, theorem	12 min
How a proof is built	12 min
Converses	8 min
Homework	5 min

LEARNING OBJECTIVES

- Name the undefined terms of geometry and say why they cannot be defined.
- Distinguish an axiom, a definition and a theorem.
- Describe the structure of a deductive proof.
- Give an example of a statement whose converse is false.

TEXTBOOK

National	pp. 3–8
Cambridge	Core & Extended, pp. 246–248
Workbook	Discussion sheet L1

01 Explanation

Three words we refuse to define

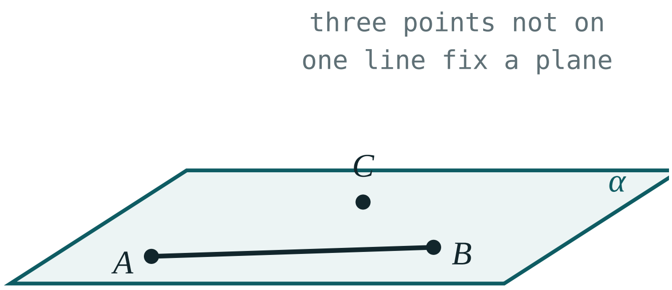
Every definition explains a new word using older words. Follow the chain back far enough and it must stop somewhere, or it runs in a circle. Geometry stops at three:

> THE UNDEFINED TERMS
>
> **point** · **line** · **plane**. They are not explained. They are described by how they behave, and that behaviour is set out by the axioms.

“A point is that which has no part” — Euclid's own attempt — defines “point” using “part”, which he never defines either. Modern geometry is honest about the stopping place.

Three kinds of statement

KIND	STATUS	EXAMPLE
axiom	accepted without proof	Through any two points passes exactly one line
definition	a name given to a described object	A trapezium is a quadrilateral with exactly one pair of parallel sides
theorem	proved from axioms, definitions and earlier theorems	The angles of a triangle sum to $180°$



Three axioms in one picture: two points fix a line, three non-collinear points fix a plane, a line with two points in a plane lies in it.

> A DEFINITION IS NEVER TRUE OR FALSE
>
> It is a naming convention. Arguing whether “a square is a rectangle” is *true* is arguing about a definition — and the answer is simply whichever the textbook chose. In this course a square **is** a rectangle.

The shape of a proof, and the converse trap

A proof is a chain: **Given** → statements, each justified by an axiom, a definition or an earlier theorem → **therefore** the conclusion. Every line needs a reason written beside it.

The **converse** swaps the given and the conclusion. It is a different statement and needs its own proof — sometimes it is false:

STATEMENT	CONVERSE	CONVERSE TRUE?
A square has four equal sides	A shape with four equal sides is a square	no — a rhombus
Vertical angles are equal	Equal angles are vertical	no
$a^2 + b^2 = c^2 \Rightarrow$ right angle	right angle $\Rightarrow a^2 + b^2 = c^2$	yes — both are theorems

02 Worked examples

EXAMPLE 1 **Classify: "Through a point not on a line there is exactly one parallel to it."**

1. Can it be proved from the others?
   For two thousand years people tried and failed.
2. It is assumed.
   Euclid's fifth postulate.
3. Denying it gives a different, consistent geometry.

ANSWER

An axiom

EXAMPLE 2 **Write the converse of "If a quadrilateral is a parallelogram then its diagonals bisect each other", and say whether it is true.**

1. Swap given and conclusion.
   If the diagonals bisect each other then it is a parallelogram.

2. Is it provable?
   Yes — a standard Grade 8 theorem.

ANSWER

True; it is the standard test for a parallelogram

EXAMPLE 3 **Why can "line" not be defined as "the shortest path between two points"?**

1. "Shortest" needs distance.
   Distance is defined using lines.

2. The definition is circular.

ANSWER

It defines a line in terms of something already defined by lines

03 Interactive model

Ask the class for a definition of "point", then keep asking what each new word means.

Axiom, definition or theorem?

"Through two points passes exactly one line."

- axiom
- definition
- theorem
- corollary

"A rhombus is a parallelogram with four equal sides."

- axiom
- definition
- theorem
- converse

"The angles of a triangle sum to $180°$."

- axiom
- definition
- theorem
- undefined term

The undefined terms of geometry are:

- point, line, plane
- point, angle, area
- line, circle, square
- length, angle, area

The converse of a true theorem is:

- always true
- always false
- sometimes true
- meaningless

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Axiom (postulate)	Aksioma	Аксиома
Undefined term	Ta’riflanmaydigan tushuncha	Неопределяемое понятие
Definition	Ta’rif	Определение
Theorem	Teorema	Теорема
Proof	Isbot	Доказательство
Converse	Teskari teorema	Обратная теорема
Corollary	Natija	Следствие
Deduction	Deduksiya	Дедукция
Consistent system	Zid bo‘lmagan sistema	Непротиворечивая система
Euclidean geometry	Evklid geometriyasi	Евклидова геометрия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

An axiom is:

- proved
- assumed
- a name
- a guess

A theorem must have:

- a diagram
- a proof
- a converse
- a number

Which is **not** undefined in geometry?

- point
- line
- plane
- triangle

Swapping the given and the conclusion gives the:

- corollary
- converse
- axiom
- proof

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Name the three undefined terms

   ANSWER point, line, plane

2. Is “through two points passes one line” an axiom or theorem?

   ANSWER axiom

3. Is “the angles of a triangle sum to 180°” an axiom or theorem?

   ANSWER theorem

4. What is a corollary?

   ANSWER A result following immediately from a theorem

5. Give the converse of “if it is a square it has four right angles”

   ANSWER If it has four right angles it is a square

6. Is that converse true?

   ANSWER no — a rectangle

7. Can a definition be false?

   ANSWER no — it is a naming convention

Medium · the standard question

7 PROBLEMS

1. Classify “vertically opposite angles are equal”

   ANSWER theorem

2. Classify “a trapezium has one pair of parallel sides”

   ANSWER definition

3. Write the converse of Pythagoras and say whether it is true

   ANSWER If $a^2+b^2=c^2$ the angle is right — true

4. Why must some terms be undefined?

   ANSWER To stop the chain of definitions running in a circle

5. Give a false converse of your own

   ANSWER e.g. “equal areas ⇒ congruent”

6. What does “consistent” mean of an axiom system?

   ANSWER No contradiction can be derived from it

7. Name a geometry where the parallel axiom fails

   ANSWER Spherical or hyperbolic geometry

Hard · stretch and reason

7 PROBLEMS

1. Explain why the parallel postulate cannot be proved from the others

   ANSWER Consistent geometries exist in which it is false

2. Give a statement, its converse, and show exactly one is true

   ANSWER e.g. “square ⇒ rhombus” true; converse false

3. Is “a square is a rectangle” true?

   ANSWER Yes under our definition — a naming choice

4. Prove: if two lines meet, they meet in exactly one point

   ANSWER Two common points would force one line, by the first axiom

5. What would happen if we assumed two parallels through a point?

   ANSWER Hyperbolic geometry — consistent, but not Euclidean

6. Distinguish “proof by contradiction” from a direct proof

   ANSWER It assumes the negation and derives an impossibility

7. Why is a diagram not a proof?

   ANSWER It shows one case; a proof must cover all

07 Homework

Homework — 4 tasks

Task 4 should be no more than five lines.

1. Classify five statements from Chapter I as axiom, definition or theorem.
2. Write the converse of three theorems you know and say which are true.
3. Explain, in your own words, why “point” is not defined.
4. Find one statement in everyday life whose converse is false, and write both.